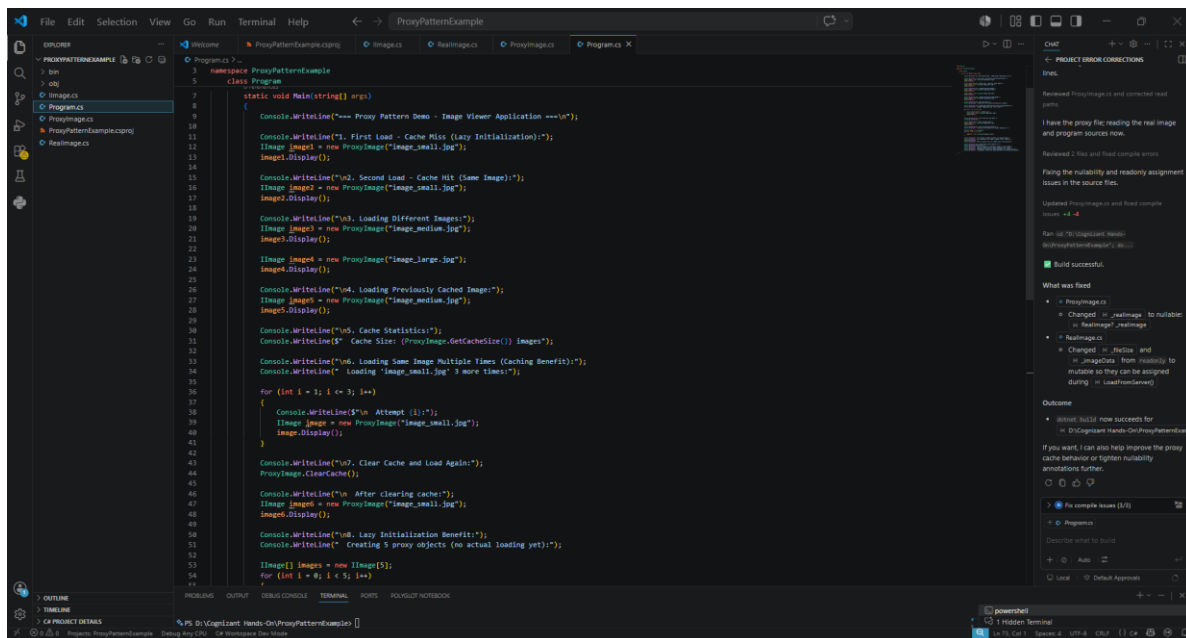


Hands-On

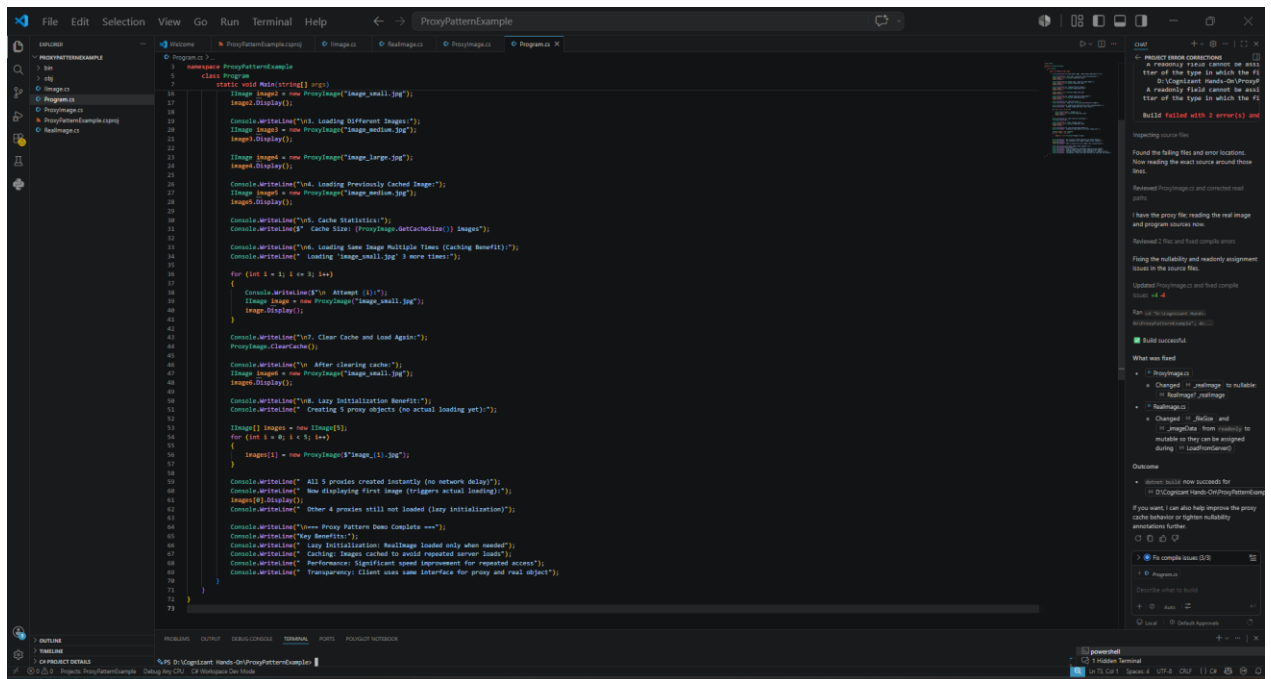
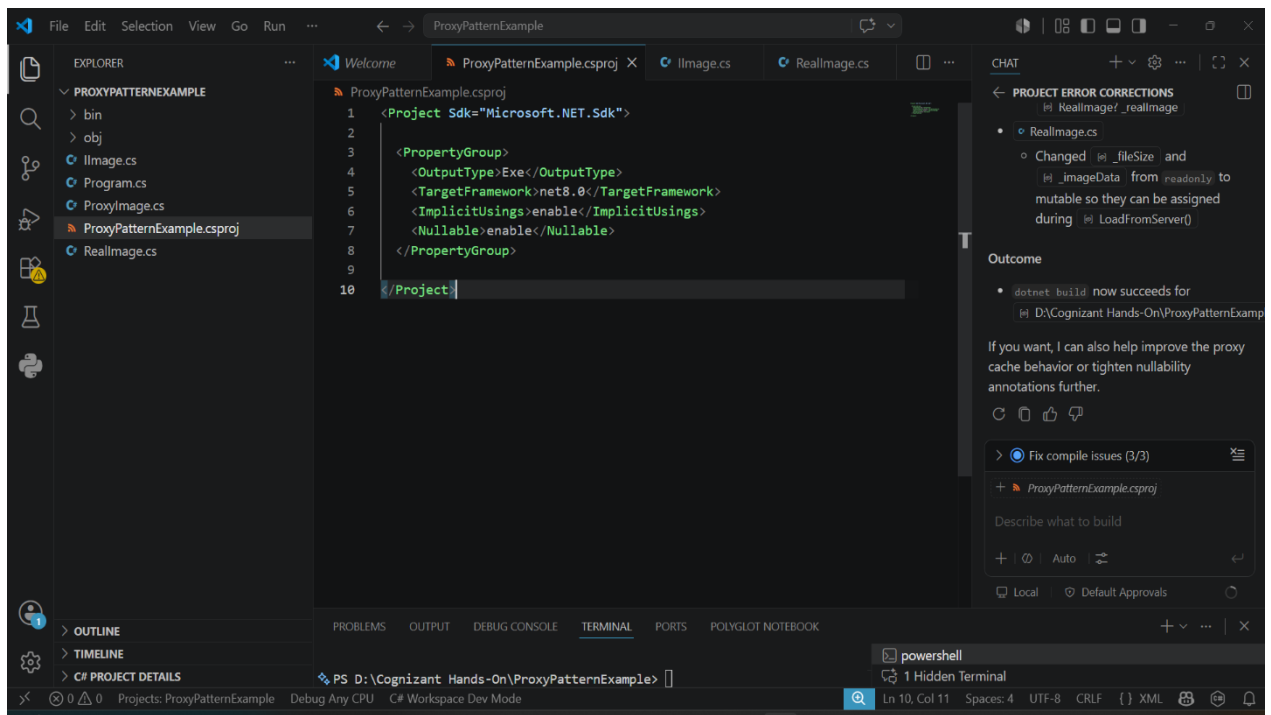
Exercise 6: Proxy Pattern Example

In this module, I have successfully implemented, executed, and tested the Proxy Pattern in .NET by creating a project called ProxyPatternExample. I developed an Image interface and a RealImage class that represented images retrieved from a remote server and included a Display() method. I created a ProxyImage class that implements the Image interface and holds a reference to the RealImage. The purpose of the ProxyImage class was to implement lazy initialization and caching so that images would only be loaded when required and reused if they were accessed multiple times. After implementing the program, I executed it and tested the loading and displaying of images through the proxy to verify that the lazy loading and caching mechanism worked as expected. This hands-on exercise helped me understand the advantages of the Proxy Pattern in improving application performance, reducing unnecessary resource consumption, and providing controlled access to resource-intensive objects in .NET applications.

Coding:



```
1 namespace ProxyPatternExample
2 {
3     class Program
4     {
5         static void Main(string[] args)
6         {
7             Console.WriteLine("--- Proxy Pattern Demo - Image Viewer Application ---");
8
9             Console.WriteLine("1. First Load - Cache Miss (Lazy Initialization):");
10            Image image1 = new ProxyImage("image_small.jpg");
11            image1.Display();
12
13            Console.WriteLine("2. Second Load - Cache Hit (Same Image):");
14            Image image2 = new ProxyImage("image_small.jpg");
15            image2.Display();
16
17            Console.WriteLine("3. Loading Different Images:");
18            Image image3 = new ProxyImage("image_medium.jpg");
19            image3.Display();
20
21            Image image4 = new ProxyImage("image_large.jpg");
22            image4.Display();
23
24            Console.WriteLine("4. Loading Previously Cached Image:");
25            Image image5 = new ProxyImage("image_medium.jpg");
26            image5.Display();
27
28            Console.WriteLine("5. Cache Statistics:");
29            Console.WriteLine($"Cache Size: {ProxyImage.GetCacheSize()} Images");
30
31            Console.WriteLine("6. Loading Same Image Multiple Times (Caching Benefit):");
32            Console.WriteLine("Loading 'image_small.jpg' 3 more times.");
33            for (int i = 1; i <= 3; i++)
34            {
35                Console.WriteLine($"on Attempt {i}:");
36                Image image = new ProxyImage("image_small.jpg");
37                image.Display();
38            }
39
40            Console.WriteLine("7. Clear Cache and Load Again:");
41            ProxyImage.ClearCache();
42
43            Console.WriteLine("8. After clearing cache:");
44            Image image6 = new ProxyImage("image_small.jpg");
45            image6.Display();
46
47            Console.WriteLine("9. Lazy Initialization Benefit:");
48            Console.WriteLine("Creating 3 proxy objects (no actual loading yet):");
49            Image[] images = new Image[3];
50            for (int i = 0; i < 3; i++)
51            {
52                images[i] = new ProxyImage($"image_{i+1}.jpg");
53            }
54        }
55    }
56 }
```



Output:

```
PS D:\Cognizant Hands-On\ProxyPatternExample> dotnet build
Restore complete (0.3s)
ProxyPatternExample net8.0 succeeded (0.3s) -> bin\Debug\net8.0\ProxyPatternExample.dll

Build succeeded in 1.0s
PS D:\Cognizant Hands-On\ProxyPatternExample> dotnet run
--- Proxy Pattern Demo - Image Viewer Application ---

1. First Load - Cache Miss (Lazy Initialization):
Proxy created for image_small.jpg (Proxy #1)
Lazy loading: image_small.jpg not loaded yet
Cache miss: Loading image_small.jpg from server
Loading image image_small.jpg from https://images.example.com...
Image image_small.jpg loaded successfully
File size: 1080 B
Displaying image: image_small.jpg
Dimensions: 800x600 pixels
Image displayed with 1 KB of data

2. Second Load - Cache Hit (Same Image):
Proxy created for image_small.jpg (Proxy #2)
Lazy loading: image_small.jpg not loaded yet
Cache hit: Loading image_small.jpg from cache
Image image_small.jpg already loaded
Displaying image: image_small.jpg
Dimensions: 800x600 pixels
Image displayed with 1 KB of data

3. Loading Different Images:
Proxy created for image_medium.jpg (Proxy #3)
Lazy loading: image_medium.jpg not loaded yet
Cache miss: Loading image_medium.jpg from server
Loading image image_medium.jpg from https://images.example.com...
Image image_medium.jpg loaded successfully
File size: 1.95 KB
Displaying image: image_medium.jpg
Dimensions: 1280x720 pixels
Image displayed with 2 KB of data
Proxy created for image_large.jpg (Proxy #4)
Lazy loading: image_large.jpg not loaded yet
Cache miss: Loading image_large.jpg from server
Loading image image_large.jpg from https://images.example.com...
Image image_large.jpg loaded successfully
File size: 4.88 KB
Displaying image: image_large.jpg
```

```
File Edit Selection View Go Run Terminal Help ProxyPatternExample
Program.cs > ... ProxyPatternExample.sproj Images.cs RealImage.cs ProxyImage.cs Programs X
3 namespace ProxyPatternExample
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK
PS D:\Cognizant Hands-On\ProxyPatternExample> dotnet run
Displaying image: image_small.jpg
Dimensions: 800x600 pixels
Image displayed with 1 KB of data

7. Clear Cache and Load Again:
Image cache cleared

After clearing cache:
Proxy created for image_small.jpg (Proxy #9)
Lazy loading: image_small.jpg not loaded yet
Cache miss: Loading image_small.jpg from server
Loading image image_small.jpg from https://images.example.com...
Image image_small.jpg loaded successfully
File size: 1000 B
Displaying image: image_small.jpg
Dimensions: 800x600 pixels
Image displayed with 1 KB of data

8. Lazy Initialization Benefit:
Creating 5 proxy objects (no actual loading yet):
Proxy created for image_0.jpg (Proxy #10)
Proxy created for image_1.jpg (Proxy #11)
Proxy created for image_2.jpg (Proxy #12)
Proxy created for image_3.jpg (Proxy #13)
Proxy created for image_4.jpg (Proxy #14)
All 5 proxies created instantly (no network delay)
Now displaying first image (triggers actual loading):
Lazy loading: image_0.jpg not loaded yet
Cache miss: Loading image_0.jpg from server
Loading image image_0.jpg from https://images.example.com...
Image image_0.jpg loaded successfully
File size: 1000 B
Displaying image: image_0.jpg
Dimensions: 800x600 pixels
Image displayed with 1 KB of data
Other 4 proxies still not loaded (lazy initialization)

=== Proxy Pattern Demo Complete ===
Key Benefits:
  Lazy Initialization: RealImage loaded only when needed
  Caching: Images cached to avoid repeated server loads
  Performance: Significant speed improvement for repeated access
  Transparency: Client uses same interface for proxy and real object
PS D:\Cognizant Hands-On\ProxyPatternExample>
```